GUJARAT TECHNOLOGICAL UNIVERSITY

Diploma in Engineering — Applied Mathematics

Unit 2: Differentiation and its Applications

Topic 2.1: Concept and Definition of Differentiation

Definition: Differentiation is a mathematical method used to calculate the exact rate at which one quantity changes in relation to another. It tells you how fast a variable moves at one precise moment.

3 Key Features:

- **The Derivative:** The final answer or formula you get after differentiating is called the derivative.
- **Notation:** It is commonly written as dy/dx, meaning 'the change in y divided by the change in x'.
- **Slope of a Curve:** Graphically, it finds the exact steepness (slope) of a curved line at a single point.

3 Major Uses:

- **Speed Monitoring:** Calculating the exact speed of a moving vehicle at any given second.
- **Optimization:** Finding the perfect conditions to get the highest efficiency out of a machine.
- **Signal Processing:** Analyzing how fast electrical voltage changes in a circuit.

Real-Life Example: Think of a car speedometer. It doesn't show your average speed for the whole trip; it uses differentiation to show your exact speed at that exact millisecond.

Topic 2.2: Working Rules (Sum, Product, Division)

Definition: These working rules are a set of standard mathematical shortcuts used to differentiate complex equations that are added, subtracted, multiplied, or divided together.

3 Key Features:

- **Sum/Difference Rule:** You can separate functions connected by a plus or minus sign and differentiate them one by one.
- **Product Rule:** Used when two changing terms are multiplied together (often called the $u \cdot v$ rule).
- **Quotient Rule:** Used when one changing term is divided by another (the fraction rule).

3 Major Uses:

- **Combining Systems:** Solving engineering equations where multiple forces act on an object at once.
- **Fluid Dynamics:** Calculating changes in fluid flow when pressure and pipe width vary together.
- **Economic Forecasting:** Tracking revenue changes when both production volume and tax rates shift.

Real-Life Example: Think of a water rocket. Its weight changes as it loses water, and its speed changes as it climbs. To find its rate of acceleration, engineers use the quotient rule to manage both changing factors.

Topic 2.3: Chain Rule

Definition: The Chain Rule is a formula used to differentiate 'nested' or composite functions. It is used when one variable depends on a second variable, which in turn depends on a third variable.

3 Key Features:

- **Function Inside a Function:** It handles equations like a set of nesting Russian dolls (e.g., $y = (3x+2)^5$).
- **Layered Step Process:** You differentiate the outer layer first while keeping the inner layer untouched, then multiply by the derivative of the inner layer.
- **Linked Rates:** It links multiple individual rates together like gears in a clock.

3 Major Uses:

- **Gear Design:** Calculating how the turning speed of a tiny gear impacts the main wheel of a machine.
- **Thermodynamics:** Finding how a change in engine speed affects pressure, which then affects temperature.
- **Robotics:** Determining how moving a robotic shoulder impacts the final position of the mechanical fingers.

Real-Life Example: Imagine blowing up a balloon. As you pump air, the volume grows. The changing volume forces the radius to expand, which changes the surface area. The chain rule connects all these expanding steps together.

Topic 2.4: Derivative of Implicit Functions

Definition: An implicit function is an equation where the variables x and y are tightly mixed together and cannot be easily separated into a simple '$y =$' format.

3 Key Features:

- **Mixed Terms:** Equations contain terms where x and y interact directly (like $x^2 + xy + y^2 = 9$).
- **Simultaneous Derivation:** You differentiate both sides of the equation with respect to x at the same time.
- **The dy/dx Chain:** Every time you differentiate a term containing y, you must attach a dy/dx next to it.

3 Major Uses:

- **Orbit Mapping:** Tracking the paths of satellites, which follow elliptical or complex circular loops.
- **Mechanical Cam Design:** Calculating slopes on irregular mechanical parts that do not follow simple straight paths.
- **Material Stress Analysis:** Finding the rate of change of stress in materials under complex multi-directional loads.

Real-Life Example: Think of the profile of a car tire. Its circular shape is represented by a mixed equation ($x^2 + y^2 = r^2$). Engineers use implicit differentiation to find the exact slope or contact point of the tire on the road.

Topic 2.5: Derivative of Parametric Functions

Definition: Parametric functions express variables like x and y independently using a third, common variable called a parameter (usually t for time or θ for an angle).

3 Key Features:

- **Independent Equations:** Instead of y directly connecting to x, you get two separate equations (e.g., $x = f(t)$ and $y = g(t)$).
- **The Bridge Formula:** You find the final rate of change using the ratio: $dy/dx = (dy/dt) / (dx/dt)$.
- **Path Tracking:** It isolates horizontal movement from vertical movement perfectly.

3 Major Uses:

- **Projectile Motion:** Tracking a missile or cricket ball by separating its horizontal forward distance from its vertical height over time.
- **Robotic Pathing:** Programming a drone's independent X and Y grid movements relative to a master clock.
- **CNC Machining:** Guiding automated cutting tools along smooth, pre-programmed curves.

Real-Life Example: Think of an etch-a-sketch toy or a computer mouse. When you move diagonally, your hand controls a horizontal track and a vertical track at the same time. A parametric equation calculates your exact path over time.

Topic 2.6: Logarithmic Differentiation

Definition: Logarithmic differentiation is a special trick where you apply natural logarithms (ln or log) to both sides of a complex equation to turn difficult powers or large multiplications into simple addition and subtraction problems before differentiating.

3 Key Features:

- **Power Breakdown:** Essential for equations where a variable sits in the power position (like $y = x^x$).
- **Simplification Rule:** It uses log rules to turn multiplication into addition and powers into coefficients ($\log a^b = b \log a$).
- **Two-Step Answer:** The final derivative always requires you to multiply your simplified answer back by the original function y.

3 Major Uses:

- **Complex Product Simplification:** Differentiating equations containing long strings of fractions multiplied together.
- **Growth and Decay Models:** Calculating rapid bacterial growth rates or compound interest shifts in financial engineering.
- **Acoustic and Audio Engineering:** Analyzing sound intensity levels which inherently scale on log tables (decibels).

Real-Life Example: Imagine tracking earthquake magnitudes on the Richter scale. Because the energy increases exponentially, engineers use logarithmic differentiation to cleanly find the rate of change of the destruction forces.

Topic 2.7: Successive Differentiation (Up to Second Order)

Definition: Successive differentiation is the process of differentiating a function multiple times in a row. Differentiating a function once gives the first derivative (dy/dx), and doing it again gives the second derivative (d^2y/dx^2).

3 Key Features:

- **Rate of a Rate:** The second derivative measures how fast the first rate of change itself is changing.
- **Notation Shift:** Written as d^2y/dx^2, y'', or $f''(x)$.
- **Curvature Indicator:** It mathematically reveals whether a graph is bending upwards like a cup or downwards like a dome.

3 Major Uses:

- **Acceleration Tracking:** Moving from distance $\rightarrow$ speed (1st derivative) $\rightarrow$ acceleration (2nd derivative).
- **Structural Safety:** Measuring the curvature and bending stress of structural beams under heavy loads.
- **Control Systems:** Designing self-balancing cruise control systems in modern automobiles.

Real-Life Example: When riding a roller coaster, the first derivative tells you your speed. The second derivative calculates the acceleration—the sudden G-force push you feel in your chest as the train accelerates down a drop.

Topic 2.8: Applications (Velocity, Acceleration, Maxima & Minima)

Definition: This topic puts calculus to work by using first and second derivatives to track physical motion and find the absolute highest (maximum) or lowest (minimum) points of an engineering problem.

3 Key Features:

- **Motion Physics:** Distance (*s*) differentiated with respect to time (*t*) gives velocity (*v*). Differentiating velocity gives acceleration (*a*).
- **Turning Points:** To find peak high points or low points, you set the first derivative to zero ($dy/dx = 0$).
- **The Peak Test:** If the second derivative at a turning point is negative, it's a maximum peak; if it is positive, it's a minimum valley.

3 Major Uses:

- **Cost Minimization:** Finding the exact dimensions of a container to use the least amount of metal sheet, saving production costs.
- **Efficiency Optimization:** Calculating the best engine speed to consume the minimum amount of fuel.
- **Safety Thresholds:** Finding the maximum weight capacity a bridge or crane can support before buckling.

Real-Life Example: A manufacturing company uses a profit equation. By finding the maximum point of that equation, they can tell factory managers exactly how many units to produce per day to make the absolute highest profit possible.